

HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG
KHOA CÔNG NGHỆ THÔNG TIN I



FINAL REPORT
Intelligent Business Management

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Hà Nội - 06/2026

EXECUTIVE SUMMARY

This report presents the results of building a Business Intelligence (BI) system to analyze the operational performance of VNRetail Conglomerate for the period 2021–2025. The project encompasses the complete BI lifecycle: data collection from three heterogeneous sources, data warehouse design, ETL pipeline engineering, data quality management, and interactive dashboard visualization across five thematic Tableau dashboards.

The simulated dataset comprises 14,981 sales orders, 40 products across 7 categories, distributed through 9 branches in 3 geographic regions, collected via 6 sales channels over 5 consecutive years. This data is enriched with two supplementary sources: Facebook Ads spend records 2,300 rows, 9 campaigns and daily weather observations include 5,478 records across 3 regions.

Instant noodles dominate revenue at 36.9% share but carry the lowest margin at 14.4%. Soft drinks exhibit the highest profit margin at 37.1% despite only 5.3% revenue share a prime upselling opportunity. The Northern region leads geographic performance at 40.0%, while Traditional Trade remains the dominant channel at 25.5%. The year 2022 saw the sharpest revenue growth +41.7% YoY, driven by post-pandemic demand recovery and campaign expansion; however, profit margins have been on a declining trend since 2023 due to rising raw material costs.

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CHAPTER 1: PROJECT INTRODUCTION

1.1 Background

In the increasingly competitive Fast-Moving Consumer market in Vietnam, large conglomerates like VNRetail require rapid, comprehensive, and accurate business data analysis capabilities to support strategic decision-making. However, business data is typically scattered across multiple systems like ERP, CRM, digital advertising platforms, and market data making consolidation and analysis complex and time-consuming.

Business Intelligence provides an integrated solution, enabling data from multiple sources to be consolidated into a unified Data Warehouse, from which interactive reports and dashboards serve decision-making at both operational and strategic levels.

1.2 Project Objectives

- Build a complete end-to-end BI system encompassing data collection, processing, storage, and visualization of business performance data.
- Design a Data Warehouse architecture using the Star Schema model suitable for revenue and marketing analytics.
- Build an automated ETL pipeline with a Streamlit-based UI for data validation and incremental loading.
- Develop 5 thematic Tableau dashboards covering executive overview, product & category, geography & branch, time & channel, and customer segment analysis.
- Extract valuable business insights and propose strategic recommendations based on analytical findings.

1.3 Project Scope

The project focuses on analyzing simulated internal business data of VNRetail Conglomerate over the 5-year period 2021–2025, including:

- Sales data: 14,981 orders, 40 products, 7 categories, 9 branches, 4 customer segments
- Advertising data: Facebook Ads spend across 9 campaigns, 2,300 daily records
- Supplementary data: Daily weather observations across 3 regions (5,478 records from Open-Meteo API)

The project does not cover consolidated financial data at the holding company level, competitor data, or external market benchmarks.

CHAPTER 2: METHODOLOGY AND SYSTEM ARCHITECTURE

2.1 Overall Architecture

The BI system is built on a standard Three-Tier Architecture:

- Source Layer: Raw input files includes the main sales Excel file, Facebook Ads CSV, and weather CSV from Open-Meteo API.
- Processing & Storage Layer: Python ETL pipeline, PostgreSQL 15 data warehouse running in a Docker container, organized in Star Schema design.
- Presentation Layer: Tableau Desktop connected directly to PostgreSQL via JDBC, displaying 5 interactive dashboards.

2.2 Technology Stack

Component	Technology	Version	Role
Database	PostgreSQL	15	Data Warehouse
Containerization	Docker / Docker Compose	Latest	Isolated environment
ETL Engine	Python + pandas + psycopg2	3.10 / 2.x	Extract, transform, load
Pipeline UI	Streamlit	Latest	Upload → validate → trigger ETL
Visualization	Tableau Desktop	2024	Interactive dashboards
Main Data Source	Microsoft Excel (.xlsx)	—	Raw sales data
Ad Data Source	CSV (Facebook Ads)	—	Campaign spend by day
Weather Source	CSV (Open-Meteo API)	—	Temperature, rainfall by region

2.3 Docker and Database Configuration

PostgreSQL 15 runs in Docker with the following configuration:

- Port mapping: 5434 (host) → 5432 (container), avoiding conflicts with any locally installed PostgreSQL
- Database: bi_db

- Schema staging: raw landing zone
- Schema dw: production data warehouse
- Credentials: bi_pass
- Configuration files: docker-compose.yml + .env (environment variables)

2.4 Data Warehouse Design

The warehouse follows a fact constellation with one central fact table and 11 surrounding dimension tables. Three fact tables share conformed dimensions, enabling cross-source analysis:

Table	Type	Rows	Description
fact_marketing_sales	Fact	14,981	Main fact: revenue, cost, profit, funnel metrics
fact_ad_spend	Fact	2,300	Facebook Ads spend by date × campaign
fact_weather_daily	Fact	5,478	Daily weather by region (2021–2025)
dim_date	Dimension	1,826	Full calendar: year, quarter, month, week
dim_product	Dimension	40	Products with category mapping
dim_branch	Dimension	9	Branches with region FK
dim_region	Dimension	3	North / Central / South
dim_channel	Dimension	6	Sales channels
dim_customer_segment	Dimension	4	Customer segments
dim_campaign	Dimension	9	Marketing campaigns
dim_department	Dimension	6	Business departments
dim_stage	Dimension	5	Conversion funnel stages
dim_machine	Dimension	4	Production machines
dim_information	Dimension	4	Customer inquiry types

To support weather-correlation analysis in Tableau, a pre-built view dw.v_sales_weather is created via CREATE OR REPLACE VIEW in the DDL script.

This view joins fact_marketing_sales with dim_branch to resolve region, dim_product, and fact_weather_daily, producing a single flat table that pairs each sales transaction with the corresponding weather conditions like temperature, humidity, and precipitation recorded in the same region on the same day. By materializing this join at the database level, Tableau connects to one source rather than performing multi-table joins at query time, reducing dashboard load latency and simplifying calculated field logic for analysts.

CHAPTER 3: DATASET DESCRIPTION

3.1 Dataset Overview

Metric	Value
Total orders	14,981
Products	40
Categories	7
Branches	9
Regions	3 regions: North, Central, South
Sales channels	6
Time period	2021–2025
Total revenue	99.1 B VND
Total profit	22.9 B VND
Avg profit margin	23.8%
Number of columns	37 fields
Supplementary source 1	Facebook Ads: 2,300 records, 9 campaigns
Supplementary source 2	Weather: 5,478 records, 3 regions × 5 years

3.2 Product Catalog by Category

Category	Products	Representative Products	Avg Margin
Mi an lien (Instant Noodles)	8	Mi Kokomi, Mi Omachi, Mi Tien Vua, Mi Kokomi lau thai	14.4%
Bia (Beer)	5	Bia Su Tu Trang lon, Bia Lion Gold, Bia Su Tu Vang chai	30.5%
Gia vi (Seasoning)	8	Nuoc mam Chin-Su, Nuoc mam Nam Ngu, Tuong ot Chin-Su	18.1%
Thuc pham (Food)	5	Pho mai Con Bo Cuoi, Thit hop Ponnice, Dau an Nakydaco	19.6%
Nuoc giai khat (Soft Drinks)	7	Nuoc khoang Vinh Hao, Nuoc tang luc Wake-up 247, Compact	37.1%

Category	Products	Representative Products	Avg Margin
Do uong (Beverages)	4	Ca phe Vinacafe 3in1, Ca phe Wake-up Saigon	23.6%
Snack	3	Banh gao Kokomi, Snack Omachi	26.9%

3.3 Geographic Structure

Region	Branches	Revenue Share	Key Market
Mien Bac (North)	Ha Noi, Hai Phong, Quang Ninh	40.0%	Largest consumer base, Mi an lien dominant
Mien Nam (South)	TP.HCM, Binh Duong, Can Tho	38.3%	Beer and modern trade concentration
Mien Trung (Central)	Da Nang, Hue, Nha Trang	21.7%	Highest margins, smaller volume

3.4 Customer Segments

Segment	Revenue Share	Margin	Orders
Kenh hien dai (Modern Trade)	26.8%	24.5%	3,782
Nha phan phoi (Distributor)	26.6%	22.7%	3,771
Ban le (Retail)	24.2%	24.1%	3,727
Thuong mai dien tu (E-commerce)	22.4%	23.7%	3,701

CHAPTER 4: DATA PIPELINE & QUALITY MANAGEMENT

4.1 ETL Flow

The pipeline executes sequentially across four stages, all triggered from the Streamlit UI. Stages 3 and 4 are only reached if Stage 2 produces no hard errors.

Stage	Component	Actions
1. Upload	Streamlit file uploader	User selects file; Streamlit reads it into a Pandas DataFrame in memory
2. Validate	validate() function in app.py	Returns hard errors, row errors, warnings, bad rows. If hard errors exist, pipeline stops here
3. Schema Init	sql-marketing.sql	Executed via docker exec psql creates or verifies all schemas, tables, views, indexes
4. Load	py-marketing.py	Full ETL for all 3 sources: Excel to staging to clean to dw

4.2 Schema Initialization

The DDL script is fully idempotent, it can be re-run at any time without data loss. This is by design: run_etl.sh executes it on every pipeline run, ensuring the schema is always in the correct state even after a Docker restart or a new environment setup.

The script applies different strategies for staging versus warehouse tables:

- **Staging schema:** Staging schema tables are truncated and reloaded on every ETL run. The tables themselves persist between runs, but their contents are fully overwritten each time. Staging holds raw data as TEXT with no type constraints and serves purely as a buffer zone data is transformed and promoted to the dw schema within the same run, so there is no value in preserving staging contents between executions.
- **Warehouse schema:** All tables in the warehouse is never dropped or truncated. Data accumulates across runs and is protected from accidental deletion. This applies to all 11 dimension tables and 3 fact tables.
- **Views:** The view sales_weather is always updated to the latest definition without requiring a drop.
- **Indexes:** 6 indexes on fact table foreign keys (full_date, channel_name, campaign_name, branch_name, product_name, and campaign on fact_ad_spend). Safe to re-run; no duplicate indexes are created.
- **Pre-seeded data:** dim_stage is seeded with the 5 funnel stages (Impression → Visit website → Click → Add to cart → Purchase). This means the canonical

funnel definition is restored on every run even if someone manually modifies the table.

4.3 Data Validation

The Streamlit validate function implements a structured quality gate with three tiers. It runs automatically when a file is uploaded, before any data is written to the database.

Tier 1: Hard Errors (block ETL entirely)

The system first checks whether all 9 required columns are present: OrderID, Date, Region, Product, Category, Quantity, Revenue, Cost, Profit. If any column is missing, validation stops immediately and returns without running further checks. The ETL button is disabled until the file is corrected and re-uploaded. This is the only check that terminates the entire pipeline.

Tier 2: Row-Level Errors (filterable, do not block ETL)

If Tier 1 passes, the system scans every row for the following 7 conditions. Each failing row index is added to a bad_rows set a row can trigger multiple checks but is only counted once:

Check	Condition flagged
Date format	Date is not parseable as YYYY-MM-DD
Null in required field	Any of the 9 required columns is empty
Revenue non-numeric	Revenue contains text or special characters
Revenue ≤ 0	Revenue is zero or negative
Quantity ≤ 0	Quantity is zero or negative
Category invalid	Value not in: Bìa, Gia vị, Mì ăn liền, Nước giải khát, Snack, Thực phẩm, Đồ uống
Region invalid	Value not in: Miền Bắc, Miền Trung, Miền Nam

The UI displays a summary card showing total records, bad record count, clean record count, and a detailed error table with the error type, example values, and row numbers.

Tier 3: Warnings (advisory only, no rows excluded)

Three additional checks are run but do not add rows to bad_rows. They surface as informational alerts only:

- CustomerSegment value not in the 4 valid segments
- Revenue < Cost — order is loss-making
- Duplicate OrderID — same order appears more than once

Auto-Filter and Error Export

When Tier 2 errors exist but Tier 1 passes, the UI presents two options side by side:

- **Auto-filter checkbox:** removes all rows in `bad_rows`, saves the cleaned DataFrame back and allows ETL to proceed with only the valid rows.
- **Download error report:** exports the bad rows as a timestamped file so the user can fix them manually and re-upload.

The ETL script is then invoked exclusively on the validated clean data — it never reads the original uploaded file directly, only the saved clean version on disk.

CHAPTER 5: BUSINESS ANALYTICS — DASHBOARD ANALYSIS

Chapter 5 presents a comprehensive analytical examination of all 17 charts deployed across the five Tableau dashboards built for the VNRetail BI system. Rather than describing each chart in isolation, this chapter follows the analytical narrative each dashboard was designed to deliver — from executive-level KPI monitoring down to granular category-branch and channel-segment diagnostics.

For each chart, the analysis covers four dimensions:

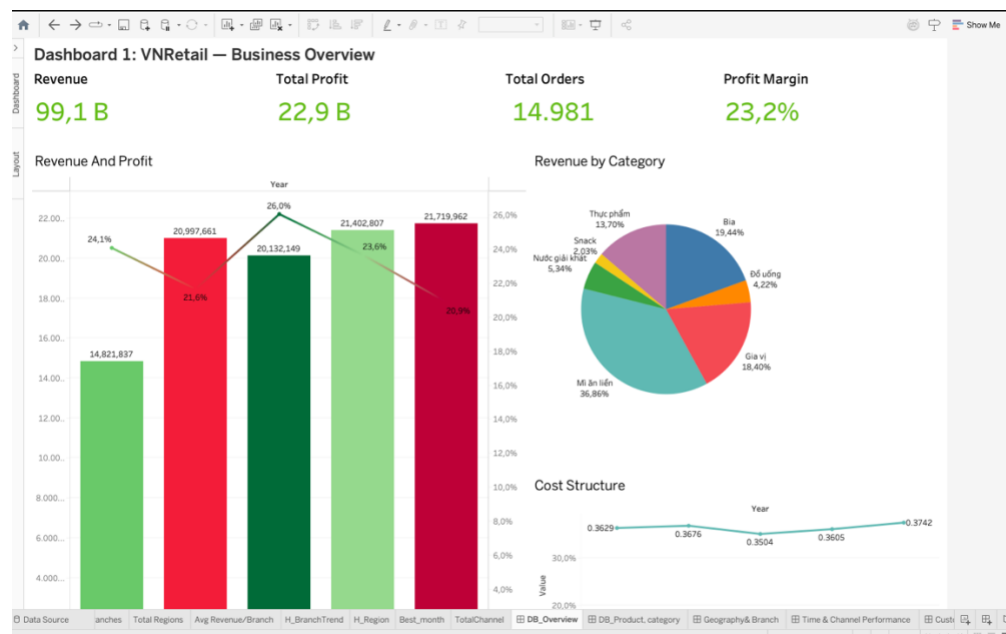
- the business question the chart was designed to answer
- the technical construction in Tableau including chart type, calculated fields, and data source configuration
- the actual results observed from the 2021–2025 dataset
- the business implications and recommendations that follow from those results.

The five dashboards progress from macro to micro:

- Dashboard 1 provides the executive overview for C-level decision-making
- Dashboard 2 drills into product and category performance
- Dashboard 3 examines geographic and branch-level distribution
- Dashboard 4 analyzes time-series trends, seasonal patterns, and sales channel efficiency; Dashboard 5 evaluates customer segment behavior and marketing ROI.

All monetary values in this chapter are expressed in VND $\times 1,000$ as stored in the data warehouse. KPI figures are displayed in billions by dividing by 1,000,000 for readability. Profit margin is calculated as $\text{Profit} / \text{Revenue} \times 100$ at the transaction level and aggregated using weighted averages, not simple averages of individual margin values.

5.1 Dashboard 1: Business Overview



Dashboard 1 serves as the executive entry point to the VNRetail BI system — a single-page summary designed for C-suite and senior management who need an immediate, high-confidence picture of five-year business performance without navigating detailed drill-downs. The dashboard answers three fundamental questions a business leader asks first: How large is the business? Which products and categories drive the most value? and Is the overall margin structure healthy

The dashboard combines four KPI cards

- Total Revenue
- Total Profit
- Average Profit Margin
- Total Orders

With three analytical charts:

- Revenue-by-category breakdown,
- Top product ranking,
- Profit margin trend over time.

Together, these seven components provide a complete performance snapshot in a single view, enabling executives to identify the scale of the business, the composition of revenue, the profitability of each category, and the trajectory of margin health across the 2021–2025 period all before drilling into any subordinate dashboard.

5.1.1. Chart Revenue and Profit Margin by Year (Dual-Axis)

Revenue And Profit Margin by Year



Chart Description

H1 is a dual-axis combination chart: vertical bar (left Y-axis) represents total revenue by year; a line overlay (right Y-axis) represents average profit margin (%). Bar colors are encoded by profit margin using a Red-Green Diverging color scheme: green bars indicate higher-margin years, red bars indicate lower-margin years. This encoding allows simultaneous reading of both revenue scale and profitability quality.

Results and Analysis

Year	Revenue (B)	YoY Growth	Profit Margin	Assessment
2021	14.8 B	—	25.3%	Base year — new product launches, lower volume
2022	21.0 B	+41.7%	24.9%	Peak growth — post-pandemic demand recovery
2023	20.1 B	-4.1%	24.5%	Slight contraction — macro headwinds
2024	21.4 B	+6.3%	22.6%	Recovery — but margin compression begins
2025	21.7 B	+1.5%	22.7%	Growth slowing, margin stabilizing at lower level

Revenue growth and Margin divergence: While total revenue grew +46.5% from 2021 to 2025 (14.8B → 21.7B), profit margin fell 2.6 percentage points from 25.3% to 22.7%. This divergence between top-line growth and margin compression is the most critical trend in the entire 5-year period. The 2022 spike (+41.7%) was exceptional, driven by post-pandemic recovery rather than structural improvement. From 2023 onward, slower growth and rising costs explain the sustained margin decline. The 2022 growth of +41.7% should not be used as a baseline for future planning. It represents a one-time post-pandemic demand surge. Sustainable organic growth for this business is estimated at 3–7% per year based on 2023–2025 trends.

5.1.2 Chart Revenue Structure by Category

Revenue by Category

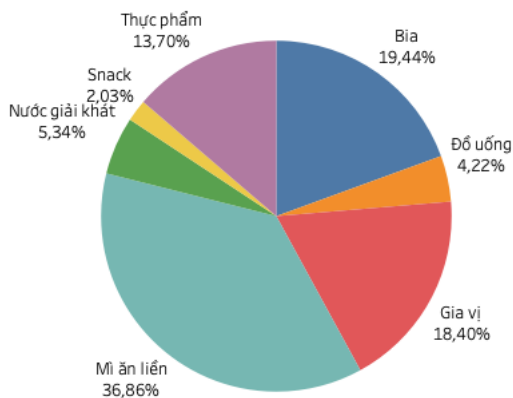


Chart Description

This is a pie chart displaying the percentage share of total 5-year revenue contributed by each of the 7 product categories. Each slice is labeled with category name and percentage. This chart is positioned to the right of H1 on Dashboard 1, providing instant visibility into the product revenue mix.

Results and Analysis

Category	Revenue (B)	Revenue Share	Margin	Category Type
Mi an lien (Instant Noodles)	36.5 B	36.9%	14.4%	Volume driver low margin
Bia (Beer)	19.3 B	19.4%	30.5%	High value, strong margin
Gia vi (Seasoning)	18.2 B	18.4%	18.1%	Stable ,moderate margin
Thuc pham (Food)	13.6 B	13.7%	19.6%	Diversified, average margin
Nuoc giai khat (Soft Drinks)	5.3 B	5.3%	37.1%	High margin, underdeveloped
Do uong (Beverages)	4.2 B	4.2%	23.6%	Niche, average margin
Snack	2.0 B	2.0%	26.9%	Small, above-average margin

Strategic Implication: Category Paradox: The pie chart reveals a classic FMCG paradox: Mi an lien, which contributes 36.9% of revenue more than the next two categories combined delivers only 14.4% margin. Conversely, Nuoc giai khat, the 5th largest category at 5.3% revenue share, delivers the highest margin at 37.1%. A 1% revenue shift from Mi an lien to Nuoc giai khat would generate approximately 225 million VND in additional profit without any top-line growth.

5.1.3 Chart Cost Structure

Cost Structure

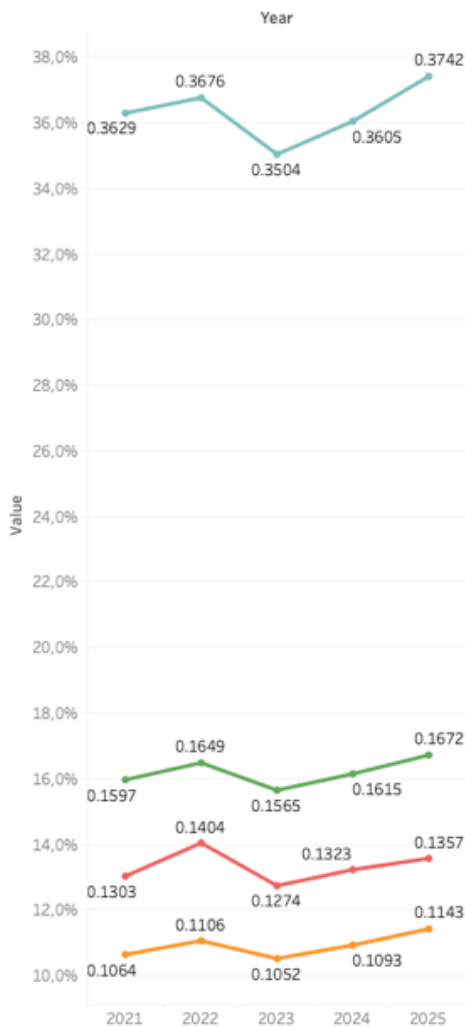


Chart Description

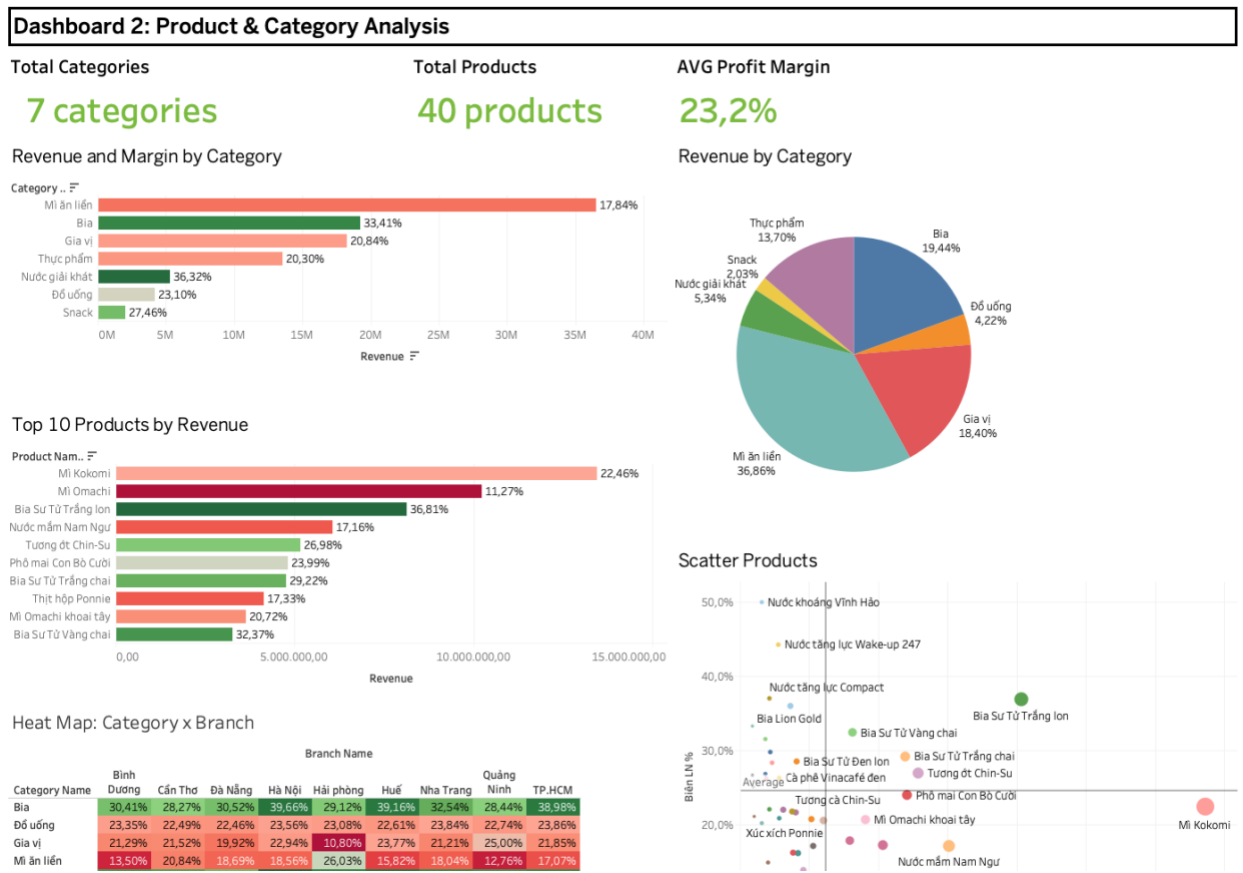
This is a pie chart decomposing total cost into four components: Raw Material Cost, Labor Cost, Logistics Cost, and Marketing Cost. This chart completes the overview story: Dashboard 1 now shows not only where revenue comes from but also where costs go.

Results and Analysis

Cost Component	Amount (B)	Percentage of Total Cost	Trend from 2021 to 2025
Raw Material Cost	36.0 B	47.2%	Revenue increased 36.3% to 37.4%
Labor Cost	16.1 B	21.1%	Revenue increased from 16.0% to 16.7%
Marketing Cost	13.2 B	17.3%	Revenue increased from 13.0% to 13.6% is the fastest growth
Logistics Cost	10.8 B	14.2%	Revenue increased smallest from 10.6% to 11.4%

Primary Cost Driver Identified: Raw materials account for nearly half (47.2%) of all costs, and their share of revenue has been increasing from 36.3% in 2021 to 37.4% in 2025. This is the single largest driver of margin compression observed in H1. Marketing costs, while smaller in absolute terms, are growing fastest proportionally (+0.6pp over 5 years), suggesting increasing competitive pressure and diminishing returns from traditional spend.

5.2 Dashboard 2: Product & Category Analysis



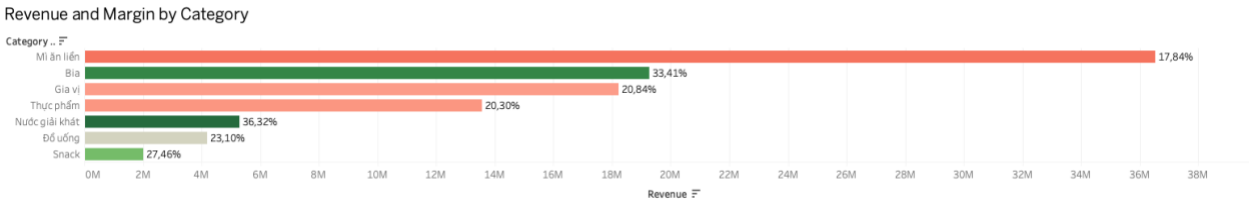


Chart Description

This horizontal bar chart ranking 7 categories by total revenue in descending order. Bar fill color is encoded by average profit margin using a Red-Green Diverging color scheme centered at the overall average 23.8%. Green bars indicate above-average margin; red bars indicate below-average margin. This dual encoding allows simultaneous assessment of both revenue scale and profitability quality in a single glance.

Results and Analysis

Rank	Category	Revenue (B)	Share	Margin	Color Signal
1	Mi an lien	36.5 B	36.9%	14.4%	Red below avg
2	Bia	19.3 B	19.4%	30.5%	Green above avg
3	Gia vi	18.2 B	18.4%	18.1%	Orange below avg
4	Thuc pham	13.6 B	13.7%	19.6%	Neutral
5	Nuoc giai khat	5.3 B	5.3%	37.1%	Dark green highest
6	Do uong	4.2 B	4.2%	23.6%	Neutral
7	Snack	2.0 B	2.0%	26.9%	Light green

Portfolio Imbalance Alert: The chart exposes a dangerous portfolio concentration: the single category with the lowest margin (Mi an lien at 14.4%) accounts for 36.9% of all revenue. Meanwhile, the two highest-margin categories (Nuoc giai khat 37.1% and Bia 30.5%) together represent only 24.7% of revenue. A rebalancing strategy shifting marketing investment toward Beer and Soft Drinks would improve overall margin without requiring revenue growth.

5.2.2 Chart Top 10 Products by Revenue

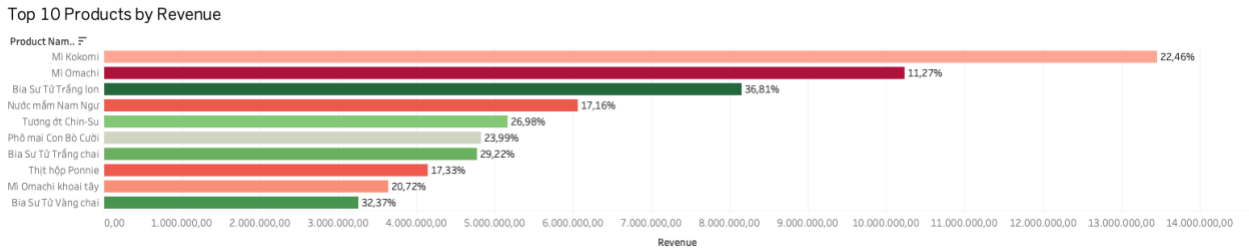


Chart Description

H4 applies a Top 10 filter by SUM of Revenue to display the highest-revenue products as a horizontal bar chart. Margin values are shown as labels on each bar, enabling product-level profitability visibility alongside revenue ranking.

Results and Analysis

Rank	Product	Revenue (B)	Margin	Assessment
1	Mi Kokomi	13.5 B	12.2%	Volume leader, margin below average
2	Mi Omachi	10.2 B	3.1%	Near breakeven, urgent review needed
3	Bia Su Tu Trang lon	8.1 B	28.7%	High revenue and strong margin
4	Nuoc mam Nam Ngu	6.1 B	11.0%	High revenue, weak margin
5	Tuong ot Chin-Su	5.2 B	23.6%	Balanced at average margin
6	Pho mai Con Bo Cuoi	4.8 B	23.0%	Premium product, healthy margin
7	Bia Su Tu Trang chai	4.8 B	27.9%	Strong performer
8	Thit hop Ponnice	4.1 B	17.5%	Neutral
9	Mi Omachi khoai tay	3.6 B	21.0%	Near average
10	Bia Su Tu Vang chai	3.3 B	33.0%	Best margin in top 10

Mi Omachi ranked 2nd in revenue at 10.2B, 3.1% margin is effectively at breakeven. At this margin, any increase in raw material or logistics costs would push the product into a loss position. This requires immediate review of pricing, cost structure, or product rationalization.

Bia Su Tu Portfolio: Three Beer products appear in the top 10 (ranks 3, 7, 10) with margins of 28.7%, 27.9%, and 33.0% respectively. Combined revenue of these three products is 16.2B, representing 16.4% of total portfolio revenue with margins nearly double the company average. This cluster is the single strongest argument for increasing investment in the Beer category.

5.2.3. Chart Product BCG Matrix

Scatter Products



Chart Description

This is a scatter chart plotting all 40 products by Revenue vs. Profit Margin, with bubble size proportional to revenue. Two reference lines divide the chart into quadrants at the average revenue 2.48B and average margin 23.6%, creating a BCG-style portfolio matrix. Points are colored by category for grouping insight.

BCG Quadrant Classification

Quadrant	Criteria	Count	Key Products	Strategy
Stars	Revenue $\geq$ avg AND Margin $\geq$ avg	4	Bia Su Tu Trang lon, Bia Su Tu Trang chai, Bia Su Tu Vang chai, Tuong ot Chin-Su	Invest and expand aggressively
Cash Cows	Revenue $\geq$ avg AND Margin $<$ avg	7	Mi Kokomi, Mi Omachi, Nuoc mam Nam Ngu, Pho mai Con Bo Cuoi, Thit hop Ponnies, Tuong ot, Mi Omachi variants	Optimize costs, harvest cash, monitor margin
Question Marks	Revenue $<$ avg AND Margin $\geq$ avg	14	Nuoc khoang Vinh Hao (48.3%), Wake-up 247 (44.2%), Nuoc tang luc Compact (36.0%), Bia Lion Gold (35.9%)	Push distribution, increase volume to move to Stars
Dogs	Revenue $<$ avg AND Margin $<$ avg	15	Muoi i-ot Vinh Hao (11%), Mien Kokomi (15%), Dau an Nakydaco (16%)	Rationalize consider discontinuation or relaunch

Priority Action: Develop Question Marks: 14 products are classified as Question Marks they already demonstrate strong margins some above 40–48% but lack the distribution scale to generate significant revenue. Nuoc khoang Vinh Hao at 48.3% margin and Nuoc tang luc Wake-up 247 at 44.2% margin are the clearest opportunities. Increasing their sales channel coverage and marketing investment should be the primary portfolio development priority.

5.2.4 Chart Profit Margin Heatmap: Category with Branch

Profit Margin HeatMap: Category x Branch

Category Name	Branch Name								
	Binh Duong	Cần Thơ	Đà Nẵng	Hà Nội	Hải phòng	Huế	Nha Trang	Quảng Ninh	TP.HCM
Bia	30,41%	28,27%	30,52%	39,66%	29,12%	39,16%	32,54%	28,44%	38,98%
Đồ uống	23,35%	22,49%	22,46%	23,56%	23,08%	22,61%	23,84%	22,74%	23,86%
Gia vị	21,29%	21,52%	19,92%	22,94%	10,80%	23,77%	21,21%	25,00%	21,85%
Mì ăn liền	13,50%	20,84%	18,69%	18,56%	26,03%	15,82%	18,04%	12,76%	17,07%
Nước giải khát	35,01%	34,05%	30,82%	40,91%	37,11%	38,93%	36,70%	35,60%	36,58%
Snack	27,58%	27,85%	28,15%	29,06%	26,63%	25,17%	27,07%	27,41%	27,20%
Thực phẩm	20,12%	20,71%	20,69%	20,01%	21,10%	20,05%	19,57%	20,28%	19,65%

Chart Description

H9 uses Tableau's square mark type to build a 7×9 matrix heatmap with categories on rows and branches on columns. Cell color encodes average profit margin using a Green

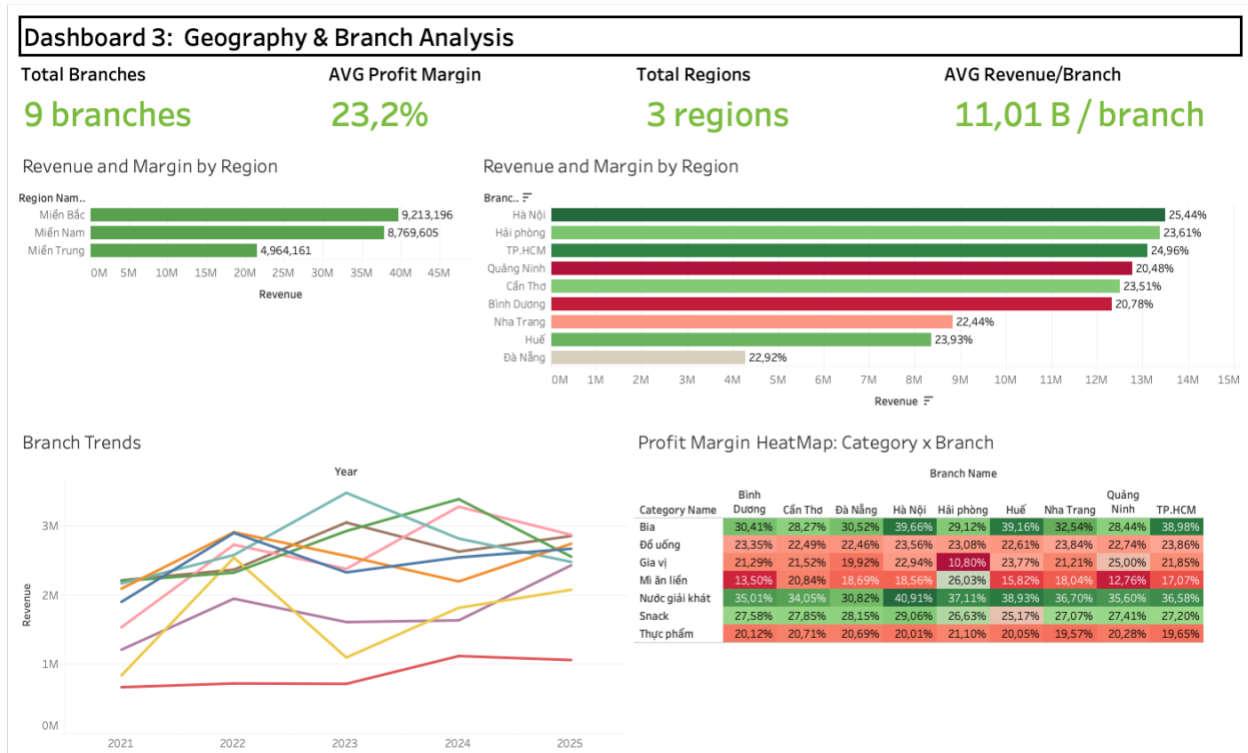
Sequential scale darker green indicates higher margin. This chart is uniquely effective at detecting geographic anomalies: cells that deviate significantly from their row category average indicate distribution, pricing, or cost issues in specific branches.

Key Observations from the Heatmap

Category	Highest Margin Branch	Lowest Margin Branch	Variance	Alert
Mi an lien	Nha Trang 19.5%	Binh Duong 10.3%	9.2pp	High, requires investigation
Nuoc giai khat	Nha Trang 41.6%	Da Nang 30.8%	10.8pp	High, logistics cost differential
Gia vi	Hue 21.2%	Hai Phong 14.0%	7.2pp	Medium, pricing variance
Bia	Da Nang 32.8%	Hai Phong 30.1%	2.7pp	Normal, category consistent
Snack	All branches	All branches	<1.5pp	Very stable, no anomaly

Anomaly: Gia vi margin at Hai Phong branch is only 14.0% with 4.1 percentage points below the category average of 18.1% and the lowest of any branch-category combination for this category. This requires investigation into local distribution costs, pricing practices, or potential inventory shrinkage.

5.3 Dashboard 3: Geography & Branch Analysis



Dashboard 3 serves as the geographic intelligence layer of the VNRetail BI system, designed for regional sales directors and branch managers who need to benchmark performance across locations and identify where resources, attention, and investment should be directed. While Dashboards 1 and 2 answer what the business is selling and how profitably, Dashboard 3 answers where, which branches are leading, which are lagging, and how each branch's trajectory has evolved over the five-year period.

The dashboard covers three analytical dimensions: a horizontal bar chart ranking all 9 branches simultaneously by revenue with margin encoded in color, enabling instant identification of branches that combine high volume with high profitability versus those that generate volume at the cost of margin; a regional revenue distribution breakdown that aggregates the three geographic zones North, Central, and South revealing the structural imbalance between the mature Northern and Southern markets and the underdeveloped Central region; a multi-line trend chart tracking each branch's annual revenue from 2021 to 2025, exposing divergent growth cycles, volatility patterns, and the branches showing momentum versus those plateauing; and a profit margin heatmap cross-tabulating all 7 categories against all 9 branches, surfacing localized anomalies such as the Gia vi margin collapse at Hai Phong 10.80% that would remain invisible in any aggregated view.

Together, these charts give regional leadership a complete geographic performance map: where the business is strong, where it is underinvested, and where operational anomalies require immediate investigation.

5.3.1 Chart = Revenue and Margin by Branch

Revenue and Margin by Branch

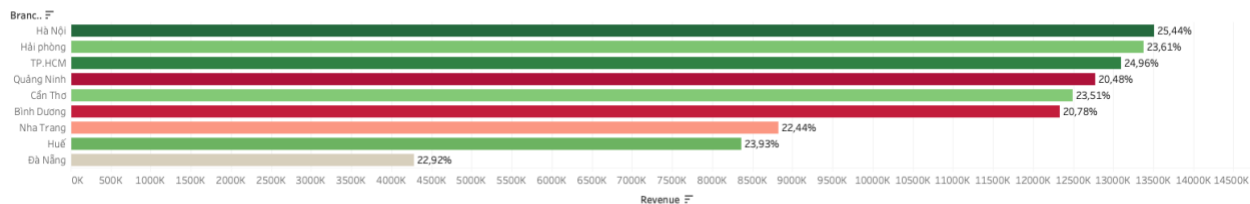


Chart Description

H6 ranks all 9 branches by total revenue using a horizontal bar chart. Bar color encodes profit margin using Red-Green Diverging, allowing simultaneous revenue and profitability comparison across all branches in a single view.

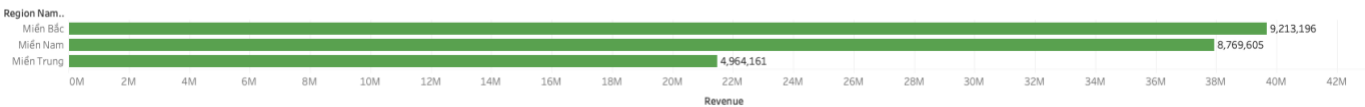
Results and Analysis

Rank	Branch	Region	Revenue (B)	Margin	Notes
1	Ha Noi	North	13.5 B	24.3%	Largest branch, above-avg margin
2	Hai Phong	North	13.4 B	23.0%	Near average, volume-focused
3	TP.HCM	South	13.1 B	24.0%	Strong metropolitan performance
4	Quang Ninh	North	12.8 B	22.3%	Slightly below average margin
5	Can Tho	South	12.5 B	24.0%	Regional hub performing well
6	Binh Duong	South	12.3 B	23.5%	Average industrial zone market
7	Nha Trang	Central	8.8 B	25.6%	Best margin smaller but profitable
8	Hue	Central	8.4 B	24.2%	Good margin, smaller market
9	Da Nang	Central	4.3 B	23.4%	Smallest significant growth potential

Central Region: Despite generating only 21.7% of total revenue, the Central region's three branches Nha Trang 25.6%, Hue 24.2%, Da Nang 23.4% all achieve above-average or average margins. Nha Trang leads all 9 branches in margin at 25.6%. This indicates the Central region serves a higher-value customer mix. Da Nang's revenue (4.3B) is less than a third of the Northern average, suggesting significant market penetration opportunity if the right category mix is deployed.

5.3.2 Regional Revenue Distribution

Revenue and Margin by Region



Region	Revenue (B)	Share	Branches	Avg Branch Revenue	Avg Margin
Mien Bac (North)	39.7 B	40.0%	Ha Noi, Hai Phong, Quang Ninh	13.2 B	23.2%
Mien Nam (South)	37.9 B	38.3%	TP.HCM, Binh Duong, Can Tho	12.6 B	23.8%
Mien Trung (Central)	21.5 B	21.7%	Da Nang, Hue, Nha Trang	7.2 B	24.4%

Strategic Gap: The North and South are nearly balanced in revenue 40.0% vs. 38.3%, but the Central region at 21.7% is significantly smaller. Notably, Central branches consistently achieve the highest average margin 24.4% vs. 23.2% North and 23.8% South. This combination of high margins and low penetration makes the Central region the highest-priority geographic expansion target.

5.3.3. Chart Branch Revenue Trends 2021–2025

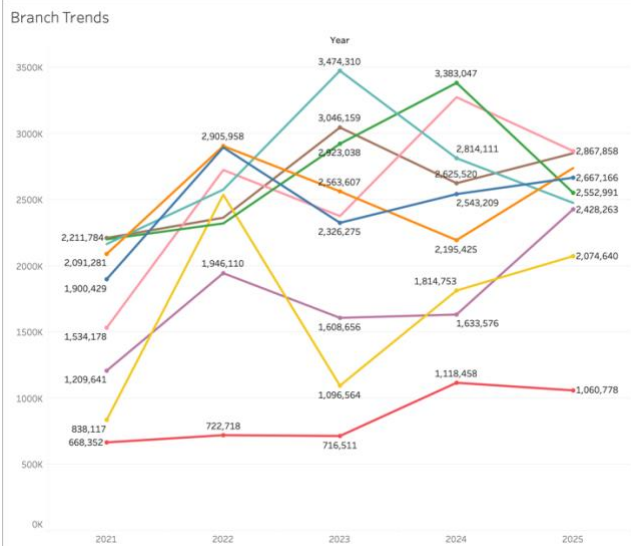


Chart Description

This multi-line chart plots annual revenue for all 9 branches over the five-year period 2021–2025, with each branch as a distinct color line. It reveals whether growth has been uniform or divergent across the network and flags branches that are accelerating, plateauing, or declining.

Results and Analysis

Branch	2021 (B)	2022 (B)	2023 (B)	2024 (B)	2025 (B)	5Y Total	Trend Pattern
Ha Noi	2.17 B	2.58	3.47	2.81	2.48	13.51	Peak 2023, moderate decline
Hai Phong	2.20	2.32	2.92	3.38	2.55	13.38	Late peak 2024 pullback
TP.HCM	2.21	2.36	3.05	2.63	2.85	13.10	Peak 2023, partial recovery 2025
Quang Ninh	1.53	2.73	2.38	3.27	2.87	12.78	Volatile, alternating high/low years
Can Tho	2.09	2.91	2.56	2.20	2.74	12.50	Peak 2022 to dip 2024 recovery
Binh Duong	1.90	2.90	2.33	2.54	2.67	12.33	Peak 2022, stabilizing
Nha Trang	1.21	1.95	1.61	1.63	2.43	8.83	Strong 2025 recovery and standout
Hue	0.84	2.54	1.10	1.81	2.07	8.36	Highly volatile 3x swing 2021 to 2022
Da Nang	0.67	0.72	0.72	1.12	1.06	4.29	Slow but steady growth still smallest

No Single Peak Year, each branch has its own cycle: Unlike a macro shock that would move all branches in sync, the branch-level data shows divergent annual peaks: Ha Noi and TP.HCM peaked in 2023, Hai Phong in 2024, Can Tho and Binh Duong in 2022. This suggests that branch performance is driven more by local market conditions, channel mix, and sales execution than by economy-wide factors. The most notable outlier is Hue: revenue tripled from 0.84B in 2021 to 2.54B in 2022 then collapsed to 1.10B in 2023 a 57% single-year drop that warrants a root-cause review. Nha Trang's 2025 surge to 2.43B up from 1.61–1.63B in 2023–2024 is the most positive trend signal in the network. Da Nang remains structurally underpenetrated at 0.67–1.12B per year across all five years, never exceeding 1.12B even in its best year (2024). At 4.29B total, it generates less than a third of the next-smallest branch Nha Trang at 8.83B. With a consistent 22–25% margin showing no operational deficiency, the constraint is sales capacity and market reach rather than profitability. A targeted branch investment plan additional sales force, local promotional budget, or distribution partnership is the logical next step.

5.3.4 Profit Margin Heatmap: Category × Branch

Profit Margin HeatMap: Category x Branch

Category Name	Branch Name								
	Bình Dương	Cần Thơ	Đà Nẵng	Hà Nội	Hải phòng	Huế	Nha Trang	Quảng Ninh	TP.HCM
Bia	30,41%	28,27%	30,52%	39,66%	29,12%	39,16%	32,54%	28,44%	38,98%
Đồ uống	23,35%	22,49%	22,46%	23,56%	23,08%	22,61%	23,84%	22,74%	23,86%
Gia vị	21,29%	21,52%	19,92%	22,94%	10,80%	23,77%	21,21%	25,00%	21,85%
Mì ăn liền	13,50%	20,84%	18,69%	18,56%	26,03%	15,82%	18,04%	12,76%	17,07%
Nước giải khát	35,01%	34,05%	30,82%	40,91%	37,11%	38,93%	36,70%	35,60%	36,58%
Snack	27,58%	27,85%	28,15%	29,06%	26,63%	25,17%	27,07%	27,41%	27,20%
Thực phẩm	20,12%	20,71%	20,69%	20,01%	21,10%	20,05%	19,57%	20,28%	19,65%

Chart Description

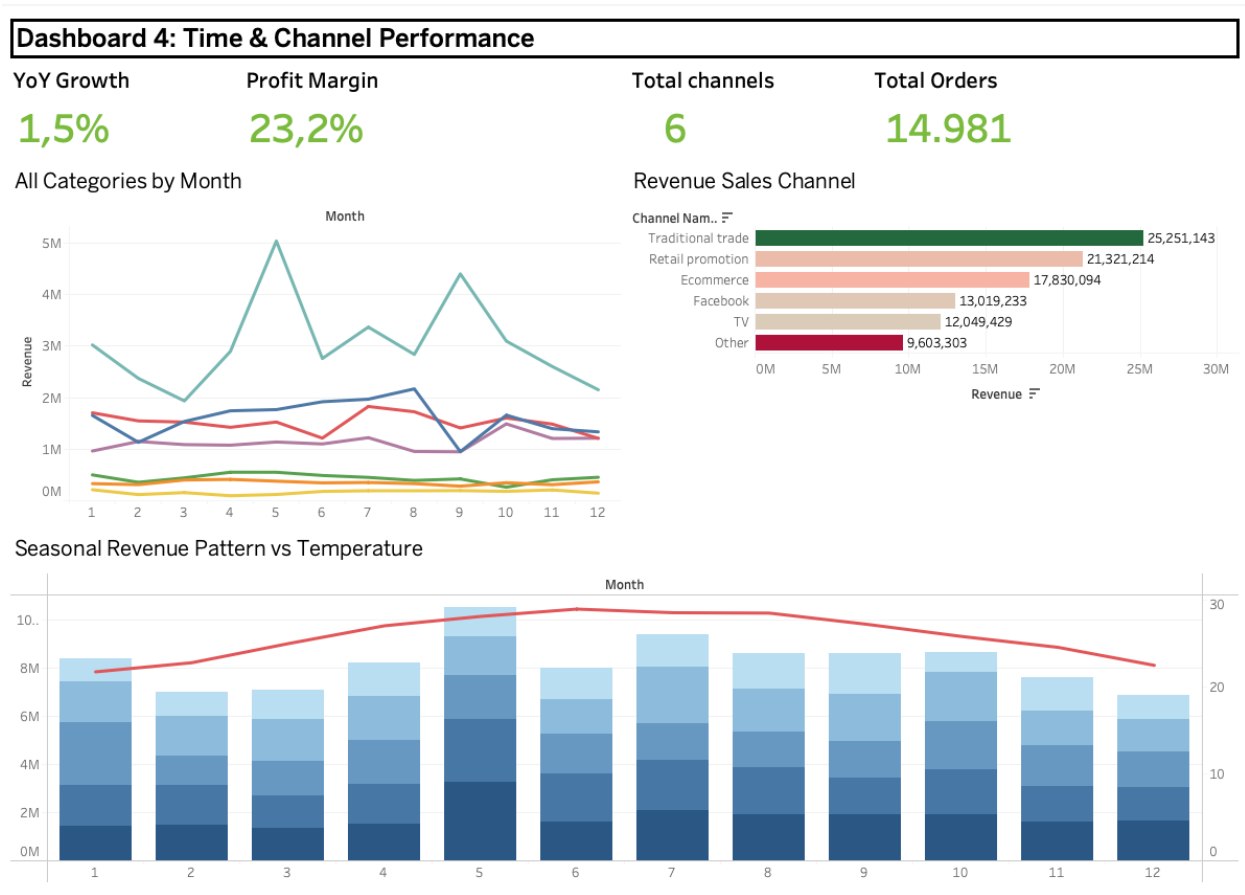
This 7×9 heatmap matrix displays profit margin for every category-branch combination. Color intensity encodes margin level, exposing localized anomalies that aggregate regional or category averages mask. A category performing well nationally may conceal a critically underperforming branch that signals structural problems in pricing, cost allocation, or distribution.

Two Critical Anomalies:

- Gia vị at Hai Phong: 10.80% versus category average 20.8% a 10.0pp gap, the largest single-cell deviation in the entire matrix. This requires immediate investigation into local distribution costs, pricing practices, or inventory shrinkage at Hai Phong.
- Mi an lien at Quang Ninh: 12.76% and at Binh Duong: 13.50%, both ~4–5pp below category average 17.8%. The pattern in industrial-zone markets (Quang Ninh, Binh Duong) suggests possible bulk discount agreements with large distributors that are not reflected in the allocated cost base, understating true margin in those branches.

Largest Strategic Margin Spread: Bia margin ranges from 28.27% Can Thon to 39.66% Ha Noi an 11.39pp spread. Ha Noi 39.66%, Hue 39.16%, and TP.HCM 38.98% all exceed 38%, while Can Tho, Hai Phong, and Quang Ninh cluster at 28–30%. Since Bia is the second-highest margin category nationally 33.4%, this urban premium effect is highly impactful. Urban markets likely sell a higher proportion of premium beer SKUs (Tiger, Heineken) versus value SKUs (333, Saigon Special) in secondary cities. Aligning SKU push strategy to this premium geography split intensifying premium Bia promotion in Ha Noi and TP.HCM while reviewing price-band positioning in Can Tho, Hai Phong, and Quang Ninh could close 2–4pp of margin gap across the network.

5.4 Dashboard 4: Time & Channel Analysis



Dashboard 4 serves as the temporal and channel intelligence layer of the VNRetail BI system, built for marketing managers and channel directors who need to understand when demand peaks and troughs, *which* sales channels deliver the best return, and whether external environmental factors such as weather measurably influence consumer purchasing behavior.

The dashboard is structured around two analytical axes. The first is time: a 7-category monthly revenue trend chart spanning the full 2021–2025 period reveals the seasonal rhythm of the business the May stockpiling spike driven by Mì ăn liền, the summer Bia surge in July–August, and the year-end moderation in November–December. These patterns are not incidental; they represent predictable demand cycles that procurement, logistics, and promotional planning teams can act on with sufficient lead time. The second axis is channel: a horizontal bar chart ranks all 6 sales channels by revenue and margin simultaneously, quantifying the shift toward e-commerce and exposing which channels carry hidden cost-to-serve disadvantages reflected in below-average margins.

Supplementing these two core charts, a weather-correlation scatter plot leverages the `dw.v_sales_weather` view to test whether temperature and precipitation levels in each

region correspond to measurable changes in category revenue providing evidence-based justification for weather-triggered promotional campaigns. Four KPI cards at the top of the dashboard anchor the analysis with period-level totals for revenue, profit, margin, and order count, giving managers immediate context before engaging with the trend detail below.

Together, Dashboard 4 equips marketing and channel teams with the analytical foundation to time campaigns more precisely, allocate channel investment more efficiently, and respond proactively to seasonal and environmental demand signals.

5.4.1 Chart Revenue Trend by Category

All Categories by Month

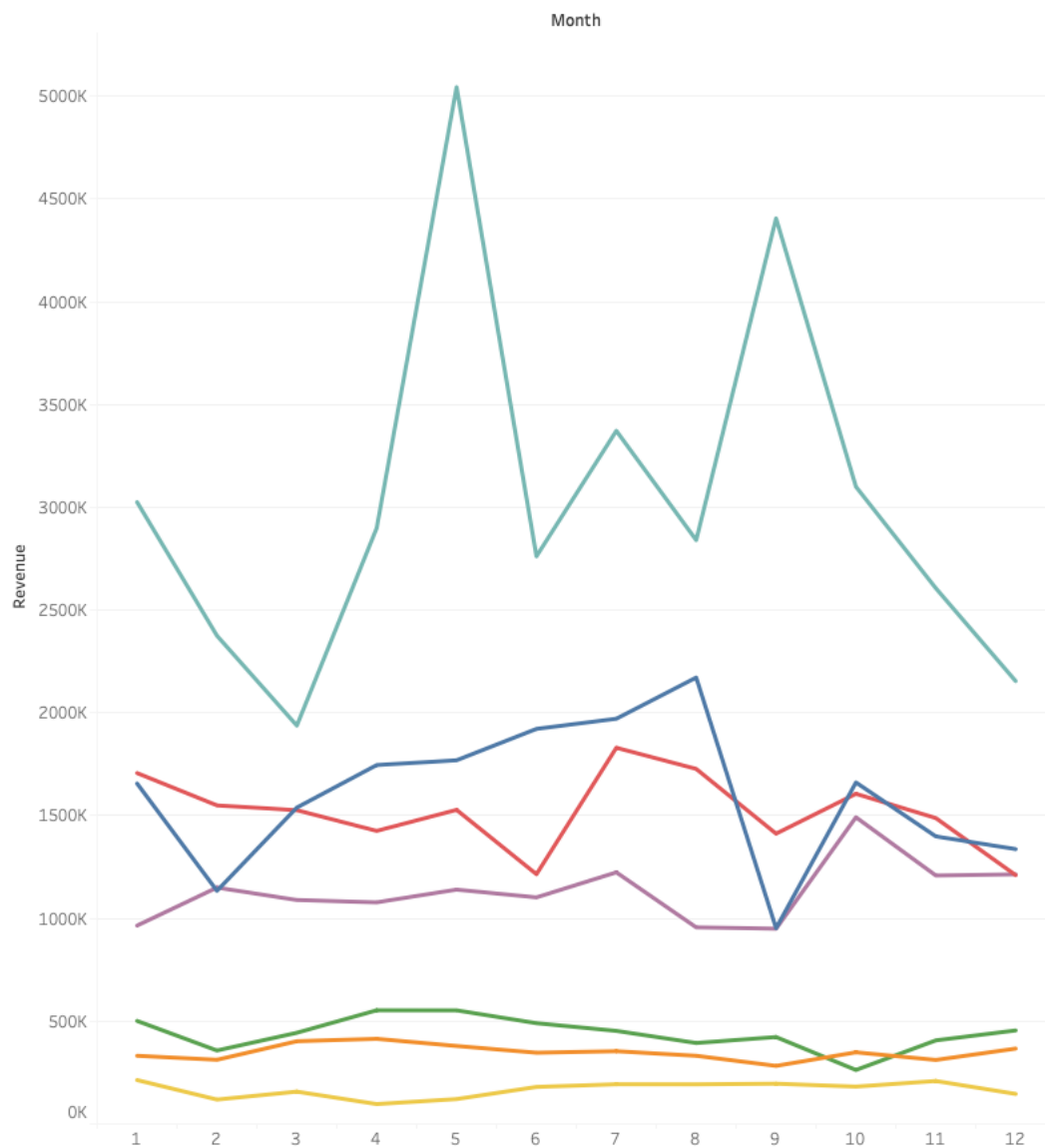


Chart Description

H7 is a multi-line time-series chart displaying monthly revenue for all 7 categories simultaneously over the full 5-year period. Each category is represented by a distinct color line. This chart is the primary tool for identifying seasonal demand patterns and category-level growth trajectories.

Key Seasonal Patterns Observed

Month	Total Revenue	Dominant Category	Seasonal Driver
January	8.4 B	Mi an lien	Post-Tet stocking
February	7.0 B	Mi an lien	Post-Tet normalization
March	7.1 B	Mi an lien	Off-season
April	8.2 B	Mi an lien	Back-to-school
May	10.5 B	Mi an lien 5.0B	Highest in May spike (stock-up season)
June	8.0 B	Mi an lien	Rainy season begins
July	9.4 B	Bia 2.0B	Beer peak because summer heat
August	8.6 B	Bia 2.2B	Beer high because hottest month
September	8.6 B	Mi an lien 4.4B	Autumn campaign
October	8.7 B	Thuc pham 1.5B	Food gifting season
November	7.6 B	Mi an lien	Pre-Tet early stock
December	6.9 B	Mi an lien	Year-end, holiday prep

The Most Important Business Pattern: May records the highest monthly revenue at 10.5B about 50% above February's low of 7.0B. This spike is driven almost entirely by Mi an lien 5.0B in May alone, likely representing distributor and retail stockpiling ahead of the summer heat season. Understanding this pattern allows procurement and logistics teams to pre-position inventory 4–6 weeks in advance, avoiding stock-outs and reducing emergency logistics costs.

Beer shows a clear summer peak in July to August versus a winter trough in November to January. For a high-margin category, this seasonal gap of ~700M/month represents significant untapped revenue. Winter marketing campaigns and bundled promotions could partially smooth this curve.

5.4.2 Chart Revenue by Sales Channel

Revenue Sales Channel

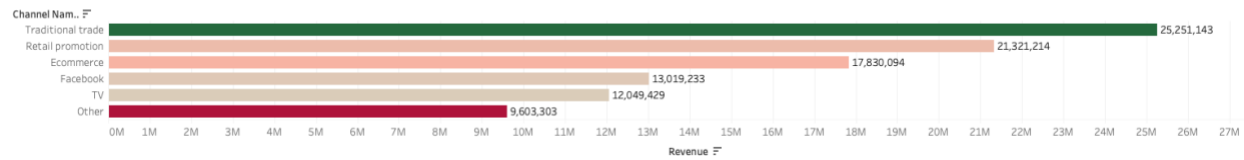


Chart Description

H8 is a horizontal bar chart ranking all 6 sales channels by total revenue. Margin values are shown as labels, enabling a direct channel efficiency comparison: which channels generate the most revenue AND the most profit per dollar of revenue.

Results and Analysis

Rank	Channel	Revenue (B)	Share	Margin	Channel Type
1	Traditional trade	25.3 B	25.5%	24.5%	Offline grocery stores, wet markets
2	Retail promotion	21.3 B	21.5%	24.4%	Offline in-store promotions
3	Ecommerce	17.8 B	18.0%	24.5%	Online e-commerce platforms
4	Facebook	13.0 B	13.1%	21.7%	Digital social commerce
5	TV	12.0 B	12.2%	23.9%	Broadcast TV shopping/infomercials
6	Other	9.6 B	9.7%	21.7%	Mixed

Digital Channels Closing the Gap: Offline channels Traditional trade and Retail promotion still dominate at 47.0% combined revenue share. However, Ecommerce alone has reached 18.0% exceeding Facebook 13.1% and TV 12.2% combined, and achieving the same 24.5% margin as Traditional trade. This suggests that online-to-offline margin parity has been achieved, removing the traditional objection to digital channel investment. Increasing Ecommerce investment now carries low risk and favorable margin profile.

Facebook and Other channels both show the lowest margin at 21.7% about 2.8 percentage points below the top channels. These channels either attract more price-sensitive customers or carry higher fulfillment costs. Before increasing spend on these channels, a deeper cost-to-serve analysis is recommended.

5.4.3 Chart Revenue vs. Temperature

Seasonal Revenue Pattern vs Temperature

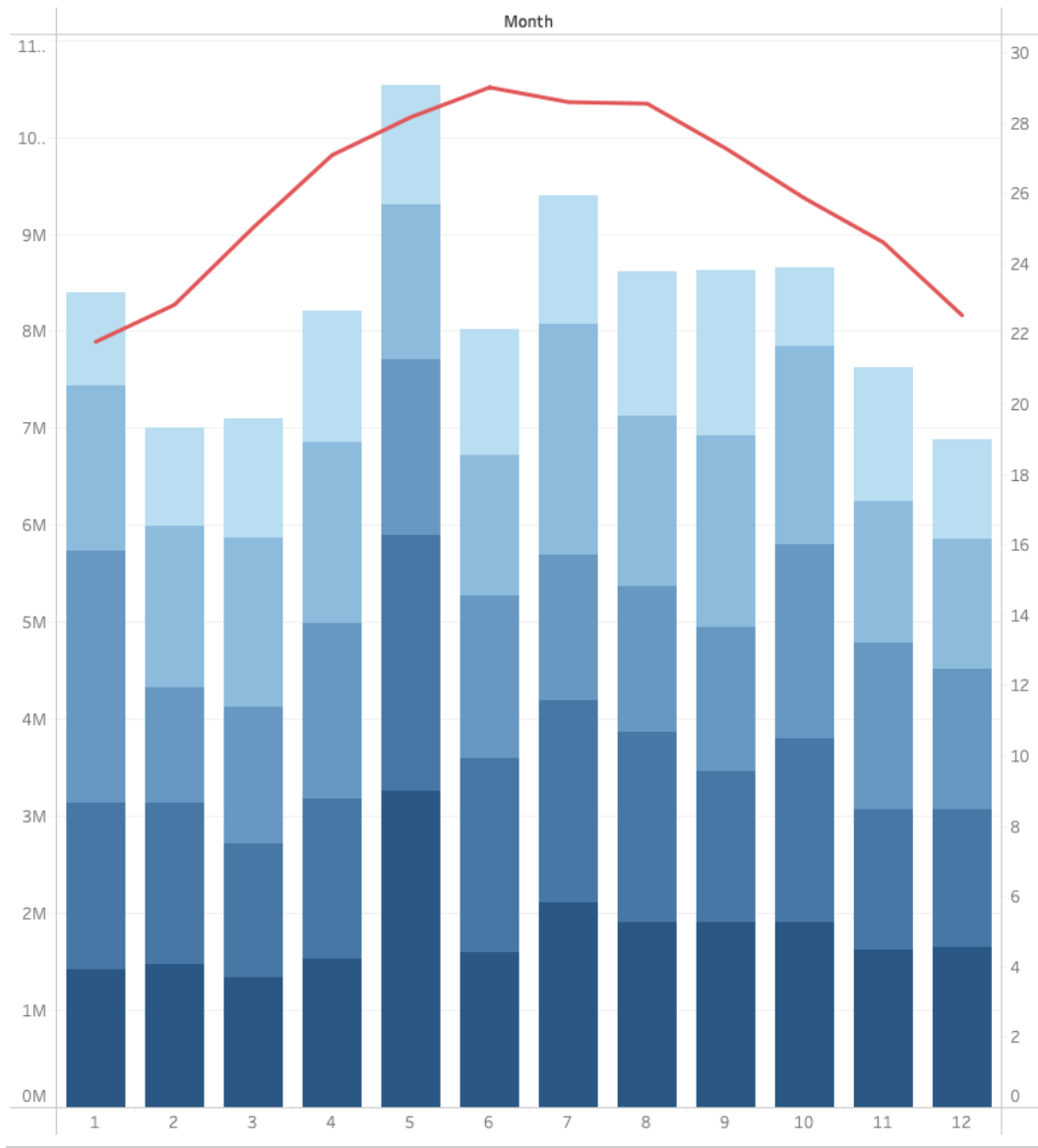


Chart Description

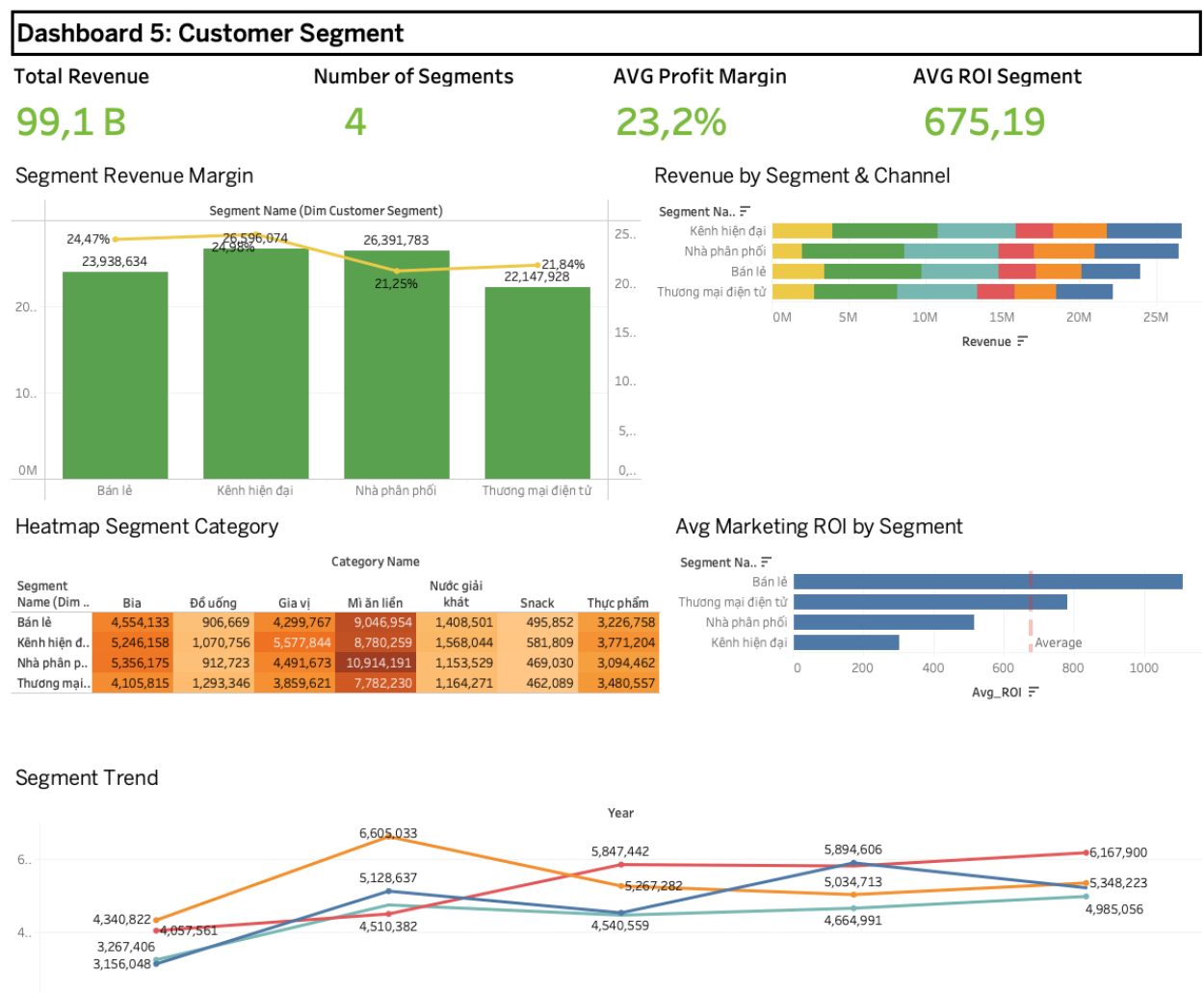
H_Weather is a dual-axis chart combining stacked bar chart monthly revenue by year, 2021–2025 stacked, left Y-axis with a line chart monthly mean temperature in °C, right Y-axis, shown in red. This chart is unique in the dashboard suite as it integrates data from two separate fact tables fact_marketing_sales and fact_weather_daily joined through the pre-built view dw.v_sales_weather.

Results and Analysis

Month	Monthly Revenue (avg)	Temp North (°C)	Temp South (°C)	Pattern
Jan–Mar	7.5 B	12–18°C	27–29°C	Cold North reduces beer; Mi an lien peaks
Apr–Jun	8.6 B	22–28°C	28–31°C	Warming drives soft drink growth
Jul–Aug	9.0 B	28–32°C	27–30°C	Heat peak, Beer and beverages peak
Sep–Oct	8.7 B	24–28°C	27–29°C	Autumn, food gifting, diverse categories
Nov–Dec	7.3 B	15–22°C	26–28°C	Cooling, Mi an lien returns, Beer declines

Temperature-Demand Correlation Confirmed: The weather chart reveals a clear positive correlation between temperature and Beer/Soft Drink revenue, and an inverse relationship between temperature and Instant Noodle revenue. Northern Vietnam's temperature range 12°C in winter to 38°C in summer creates significantly larger seasonal demand swings than the South 25–35°C year-round. Regional inventory planning should therefore apply different seasonal adjustment factors for North versus South branches.

5.5 Dashboard 5: Customer Segment Analysis



Dashboard 5 serves as the customer intelligence layer of the VNRetail BI system, designed for CRM teams and key account managers who need a granular, multi-dimensional view of how different customer segments behave, contribute, and respond to marketing investment. Where the previous dashboards answer questions about products, geographies, time, and channels at the business level, Dashboard 5 reframes every question through the lens of who which segments generate the most value, which are growing, which are being reached efficiently by marketing spend, and which channels each segment prefers.

The dashboard is built around four analytical dimensions. Revenue contribution and margin profile establish the baseline: a dual-axis chart comparing segment revenue against profit margin reveals whether high-volume segments are also high-quality segments, or whether volume is being bought at a margin cost. Growth trajectory adds the time dimension: a segment revenue trend chart across 2021–2025 identifies which segments are accelerating, plateauing, or in structural decline, enabling proactive reallocation of sales effort before problems compound. Marketing efficiency connects

advertising investment to sales outcome: by cross-referencing Facebook Ads spend from fact_ad_spend with segment-level revenue from fact_marketing_sales, the dashboard surfaces ROI differentials across campaigns and segments quantifying which marketing spend is generating returns and which is not. Channel preference closes the picture: a breakdown of each segment's revenue by sales channel reveals whether segments are shifting toward e-commerce, remaining loyal to traditional trade, or splitting across multiple touchpoints in ways that complicate fulfillment and margin management. Four KPI cards anchor the dashboard with segment-level aggregates, providing immediate orientation before managers engage with the six analytical charts beneath. Taken together, Dashboard 5 gives CRM and key account teams the evidence base to differentiate their approach by segment tailoring pricing, promotion, channel strategy, and marketing budget allocation to the specific behavioral and financial profile of each customer group rather than applying a single undifferentiated commercial strategy across all four segments.

5.5.1 Chart Revenue and Margin by Segment

Segment Revenue Margin

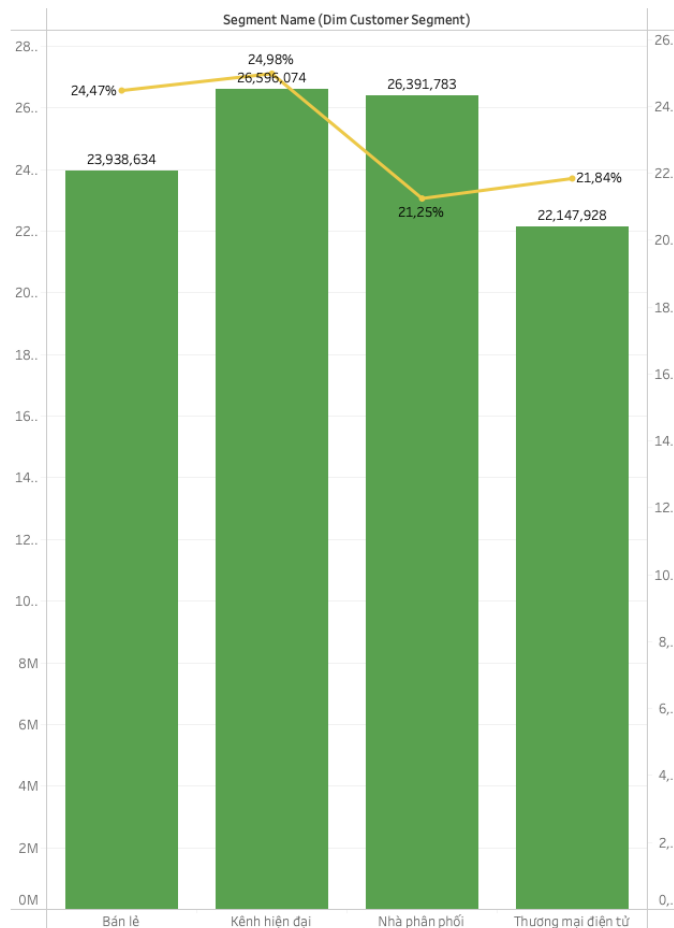


Chart Description

This chart is a combined chart displaying total revenue and average profit margin for each of the 4 customer segments. This provides a direct revenue-to-margin comparison at

the segment level, identifying which segments deliver the best combination of volume and profitability.

Results and Analysis

Segment	Revenue (B)	Share	Orders	Avg Margin	Revenue/Order
Kenh hien dai (Modern Trade)	26.6 B	26.8%	3,782	24.5%	7.0M VND
Nha phan phoi (Distributor)	26.4 B	26.6%	3,771	22.7%	7.0M VND
Ban le (Retail)	23.9 B	24.2%	3,727	24.1%	6.4M VND
Thuong mai dien tu (E-commerce)	22.1 B	22.4%	3,701	23.7%	6.0M VND

Segment Balance: Unlike the category distribution where Mi an lien dominates at 36.9%, the customer segment distribution is remarkably balanced: the gap between the largest Modern Trade 26.8% and smallest E-commerce 22.4% is only 4.4 percentage points. This diversification reduces dependency risk. However, Modern Trade commands the highest margin 24.5% making it the priority segment for growth, while Distributor margin 22.7% should be the focus for profitability improvement.

5.5.2 Chart Revenue Trend by Segment

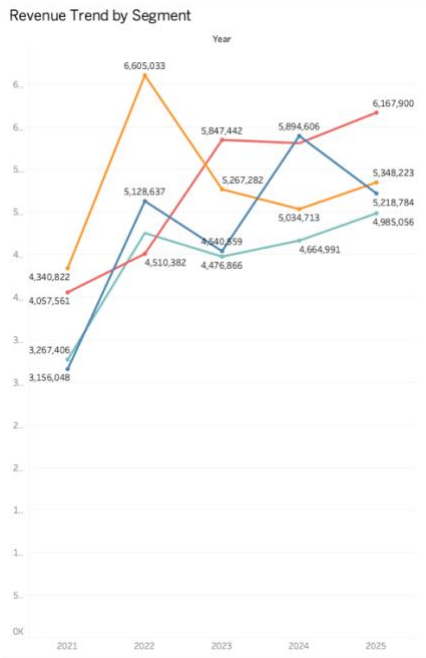


Chart Description

This is a multi-line chart showing annual revenue by customer segment from 2021 to 2025. Each segment is a distinct color line, enabling direct year-over-year comparison of segment growth rates and relative performance shifts.

Results and Analysis

Year	Modern Trade	Distributor	Retail	E-commerce	E-com Growth
2021	4.3 B	4.1 B	3.2 B	3.3 B	—
2022	6.6 B	4.5 B	5.1 B	4.8 B	+45.5%
2023	5.3 B	5.8 B	4.5 B	4.5 B	-6.3%
2024	5.0 B	5.8 B	5.9 B	4.7 B	+4.4%
2025	5.3 B	6.2 B	5.2 B	5.0 B	+6.4%

Distributor Channel Structural Growth: The Distributor segment shows the most consistent growth trajectory: revenue increased from 4.1B in 2021 to 6.2B in 2025 +51.2% total, +0.525B/year average. Notably, Distributor was the only segment to grow in 2023 the contraction year for all other segments, reaching 5.8B vs. 4.5B in 2022. This resilience indicates distributors serve as a buffer during market downturns, absorbing inventory from weaker direct channels. The Distributor segment merits priority attention for key account development.

5.5.3 Chart Revenue Heatmap: Segment and Category

Heatmap Segment Category

Segment Name (Dim Customer Se..	Category Name						
	Bia	Đồ uống	Gia vị	Mì ăn liền	Nước giải khát	Snack	Thực phẩm
Bán lẻ	4,554,133	906,669	4,299,767	9,046,954	1,408,501	495,852	3,226,758
Kênh hiện đại	5,246,158	1,070,756	5,577,844	8,780,259	1,568,044	581,809	3,771,204
Nhà phân phối	5,356,175	912,723	4,491,673	10,914,191	1,153,529	469,030	3,094,462
Thương mại điện tử	4,105,815	1,293,346	3,859,621	7,782,230	1,164,271	462,089	3,480,557

Chart Description

This is a heatmap using Tableau's Square mark type with a Green Sequential color encoding revenue intensity. Rows represent customer segments; columns represent product categories. This chart identifies which segments buy which categories disproportionately revealing cross-sell opportunities and potential over-dependencies.

Key Observations

- Mi an lien is the dominant category for all segments, but especially for the Distributor segment where it accounts for an outsized share consistent with bulk distributor purchasing patterns.

- Beer (Bia) shows stronger concentration in the Modern Trade and Retail segments, where in-store display and cold chain access drive impulse purchases.
- E-commerce segment shows relatively even distribution across categories compared to other segments, suggesting online shoppers are more exploratory and less brand-loyal.
- Soft Drinks (Nuoc giai khat) are weakest across all segments in absolute revenue supporting the case for active push campaigns to increase trial and repeat purchase.

5.5.4 Chart Revenue Segment with Channel

Revenue by Segment & Channel

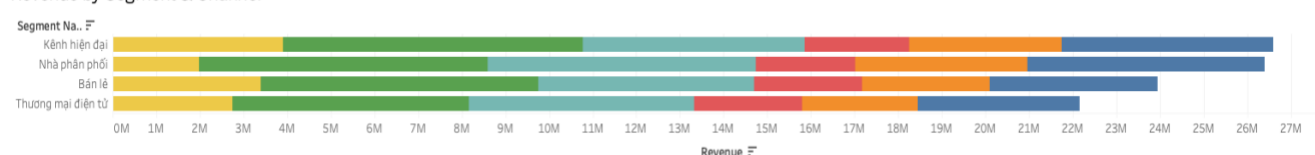


Chart Description

This stacked bar chart with customer segments on the Y-axis and revenue on the X-axis, with each bar stacked by sales channel color. This reveals the channel mix within each segment show how different customer types prefer to purchase.

Results and Analysis

Segment	Top Channel	2nd Channel	3rd Channel	Digital %	Traditional %
Modern Trade	Trad. trade 6.9B	Retail promo 5.1B	Ecommerce 4.8B	~30%	~45%
Distributor	Trad. trade 6.6B	Retail promo 6.1B	Ecommerce 5.4B	~29%	~48%
Retail	Trad. trade 6.4B	Retail promo 4.9B	Ecommerce 3.9B	~29%	~45%
E-commerce	Trad. trade 5.4B	Retail promo 5.2B	Ecommerce 3.7B	~28%	~45%

Counter-intuitive Finding: The E-commerce customer segment despite being defined by its channel still generates its highest revenue through Traditional trade 5.4B followed by Retail promotion 5.2B, with actual Ecommerce platform revenue at only 3.7B. This suggests the segment classification likely reflects the customer's business type(online retailer reselling VNRetail products rather than the purchase channel used. All four segments show similar 28–30% digital channel mix, indicating no segment has yet made a decisive shift to fully digital procurement.

5.5.5 Chart Marketing ROI by Segment

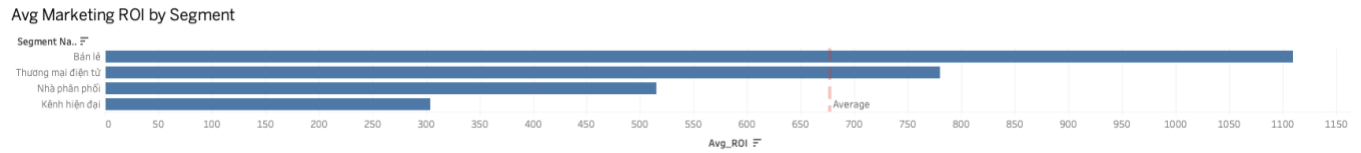


Chart Description

This is a horizontal bar chart displaying average Marketing ROI (profit / marketing cost) by customer segment, with a vertical reference line showing the overall average. Bars are sorted descending. This chart directly answers: "Which customer segments deliver the highest return on marketing investment?"

Results and Analysis

Rank	Segment	Avg Marketing ROI	vs. Average	Interpretation
1	Ban le (Retail)	1,109.5	+2.8×	Retail customers generate highest ROI per marketing spend
2	Thuong mai dien tu (E-commerce)	780.4	+1.9×	Digital native efficient marketing response
3	Nha phan phoi (Distributor)	514.9	Above avg	Bulk buyers marketing investment scales well
4	Kenh hien dai (Modern Trade)	304.0	Below avg	Highest cost-to-serve segment

Best Margin Segment Yields Lowest Marketing ROI: Modern Trade has the highest profit margin (24.5%) but the lowest marketing ROI (304.0). This is because Modern Trade requires the highest marketing investment trade promotions, display fees, key account management which drives down the return per marketing dollar. Retail, by contrast, shows 1,109.5 ROI, nearly 3.7 time higher than Modern Trade. A strategic budget reallocation moving 10% of Modern Trade marketing spend to Retail would significantly improve overall Marketing ROI without materially impacting revenue.

CHAPTER 6: CONCLUSIONS AND STRATEGIC RECOMMENDATIONS

6.1 Summary of Key Findings

Finding	Evidence	Urgency
Margin compression despite revenue growth	Margin fell 25.3% → 22.7% while revenue grew +46.5%	HIGH
Mi Omachi near-breakeven	3.1% margin on 10.2B revenue product	CRITICAL
Raw material cost escalation	RM cost share rose 36.3% → 37.4% of revenue	HIGH
Central region underdeveloped	21.7% revenue with highest margins (24.4% avg)	MEDIUM
Question Mark products with 40–48% margin	14 products with high margin but low distribution	MEDIUM
E-commerce parity with offline margins	Online margin matches traditional trade at 24.5%	OPPORTUNITY
Retail segment's marketing ROI 3.7× higher than Modern Trade	ROI 1,109 vs. 304	OPPORTUNITY
May demand spike (50% above February)	Driven by Mi an lien stockpiling	PLANNING

6.2 Strategic Recommendations

Short-Term (0–6 months)

- **CRITICAL:** Initiate an immediate cost review for Mi Omachi. At 3.1% margin, this 10.2B product is one supply shock away from a loss position. Options: price increase, ingredient substitution, or volume reduction to focus on higher-margin Mì ăn liền variants.
- **Seasonal inventory pre-positioning:** Place Mi an lien orders 6 weeks before May to avoid stock-outs during the peak month. Place Beer orders 4 weeks before July.
- **Reallocate 10–15% of Modern Trade marketing budget to Retail and E-commerce channels based on 3.7 time superior ROI.**

Medium-Term (6–18 months)

- Invest in Central region expansion: Open or expand the Da Nang branch, which currently generates only 4.3B vs. Ha Noi's 13.5B despite comparable demographics. Target Da Nang to reach 7–8B within 18 months.
- Accelerate Question Mark development: Prioritize Nuoc khoang Vinh Hao 48.3% margin, Wake-up 247 44.2% margin, and Nuoc tang luc Compact 36.0% margin with dedicated distribution pushes and targeted digital campaigns.
- Beer category investment: Three Beer products already in Top 10; Bia category margin 30.5% is $2.1\times$ that of Mi an lien. Expanding Beer SKUs and distribution especially in Northern and Central regions should be a portfolio priority.

Long-Term (18+ months)

- Category rebalancing: Reduce dependency on Mi an lien from 36.9% to below 30% of revenue over 3 years, replacing volume with higher-margin Beer, Soft Drinks, and Snack categories.
- Full e-commerce channel migration: Given proven margin parity (24.5% online = offline), build a direct-to-consumer e-commerce capability to reduce trade channel dependency and improve marketing measurement.
- Weather-responsive supply chain: Implement automated demand forecasting models that incorporate weather data (already in the data warehouse) to predict regional demand 30–60 days in advance.

6.3 Technical Recommendations

- Add real-time data ingestion: Replace monthly Excel uploads with an API connection to the ERP system, reducing data latency from monthly to daily.
- Implement data quality scoring: Extend the current validation system to generate a monthly Data Quality Score tracking trends in null rates, format violations, and duplicate rates over time.
- Tableau Server deployment: Migrate from Tableau Desktop to Tableau Server for scheduled data refresh and role-based access control, enabling branch managers to access their branch's dashboard without requiring Tableau Desktop licenses.
- Add predictive analytics: Integrate Python-based forecasting Prophet or ARIMA into the pipeline to generate 3-month revenue forecasts by category and region, surfaced directly in Dashboard 1.